October 13, 15, 2026

UNIT 2: BONDING

D. VSEPR (7.6)
--Molecules with more than one Central Atom
E. Valence Bond Theory and Orbital Hybridization (8.1, 8.2, 8.3)
F. Molecular Orbital Theory (10.3)
G. Chemistry of Selected Main Group Elements

We will have a normal class meeting on Tuesday.

For Thursday: we will not have a class meeting. I will post videos which cover the second half of these notes for Thursday's class. Please watch the videos and take notes as normal.

Generate Lewis structures for covalent molecules and ions and, coupled with VSEPR and Valence Bond Theory, determine the molecule's geometry, polarity, and bonding (bond order, bond length, strength and energy).

Use Molecular Orbital Theory to assess the bonding (bond order and strength) and magnetic properties in second-row homonuclear diatomic molecules and ions.

5. Molecules with greater than one central atom.

Example: CH_3COOH

E. Valence Bond Theory and Orbital Hybridization (8.1, 8.2, 8.3)

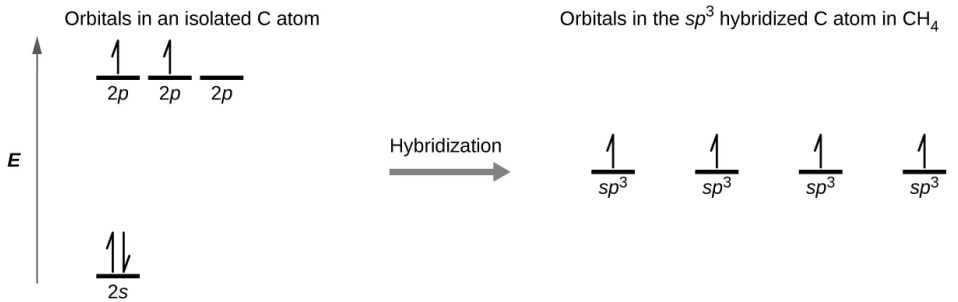
- Lewis structures and VSEPR allow us to predict geometry, but they do not tell us which atomic orbitals are involved in bonding.
- The basic principle for Valence Bond Theory is that a covalent bond forms when orbitals of two atoms overlap, and a pair of e^- is localized in the region between the atoms.
  (1) The pair of e^- has opposite spins.
  (2) Good, in-phase overlap leads to strong bonding interactions. (Phase is related to the sign of the wavefunctions of the atomic orbitals).
  (3) Orbitals can hybridize to more appropriate shapes and orientations, to maximize overlap.
  Direct overlap of the *s* and/or *p* orbitals of isolated atoms does not account for the shape of many molecules (e.g., **methane**)

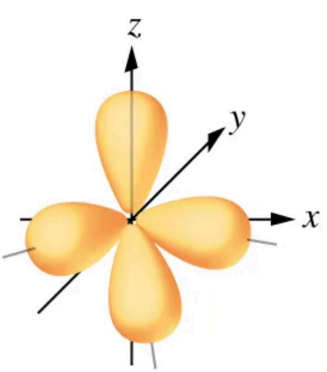
Carbon ([He]$2s^2 2p^2$) overlapping with H ($1s^1$) to form CH_4

- Linus Pauling proposed: during bonding, valence atomic orbitals (AOs) in the isolated atoms become *different* when they are in the molecule.

- Quantum-mechanical calculations show that if we mathematically "mix" certain combinations of orbitals, we form new orbitals whose spatial orientations match the observed molecular shapes.

- The process of orbital mixing is **called hybridization, and the new AOs are called hybrid orbitals.**





Hybridization is not an actual physical process; it is a hypothetical process that is invoked to explain molecular shapes that are observed experimentally.

1. Composition and Orientation of Hybrid Orbitals

- Depending on the type of atomic orbitals "mixed" and their proportions, we can get different geometries:

# of electron groups and Electron Group Geometry	Number of hybrid orbitals	Atomic Orbitals	Hybrid Orbitals Formed	Spatial Arrangement of Hybrid Orbitals

Number of hybrid orbitals corresponds to electron group geometry so we can predict hybridization from electron group geometry.

2. Examples

(a) Example: sp orbitals in $BeCl_2$

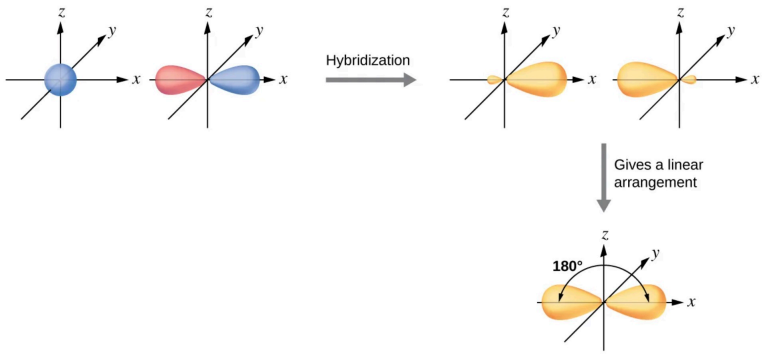
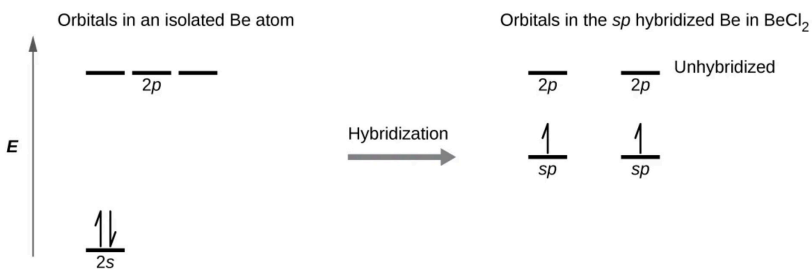


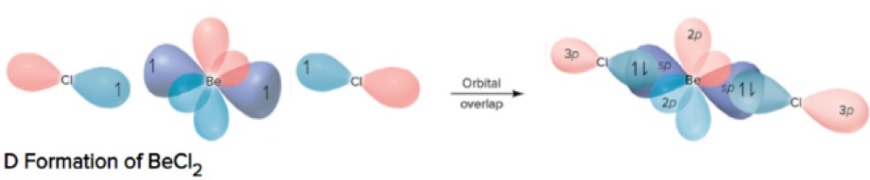
Figure 8.8 Hybridization of an *s* orbital (blue) and a *p* orbital (red) of the same atom produces two *sp* hybrid orbitals (yellow). Each hybrid orbital is oriented primarily in just one direction. Note that each *sp* orbital contains one lobe that is significantly larger than the other. The set of two *sp* orbitals are oriented at 180°, which is consistent with the geometry for two domains.

Note shapes of hybrid orbitals with one large lobe and one small lobe are different than the atomic orbitals. We usually draw only the large lobe.



Mixing one s and one p orbital gives two sp hybrid orbitals. The other two p orbitals remain unhybridized.

Figure 8.9 This orbital energy-level diagram shows the sp hybridized orbitals on Be in the linear $BeCl_2$ molecule. Each of the two sp hybrid orbitals holds one electron and is thus half filled and available for bonding via overlap with a Cl 3p orbital.



Overlap of Be and Cl orbitals to form $BeCl_2$.

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(b) sp^2 hybrid orbitals in BF_3

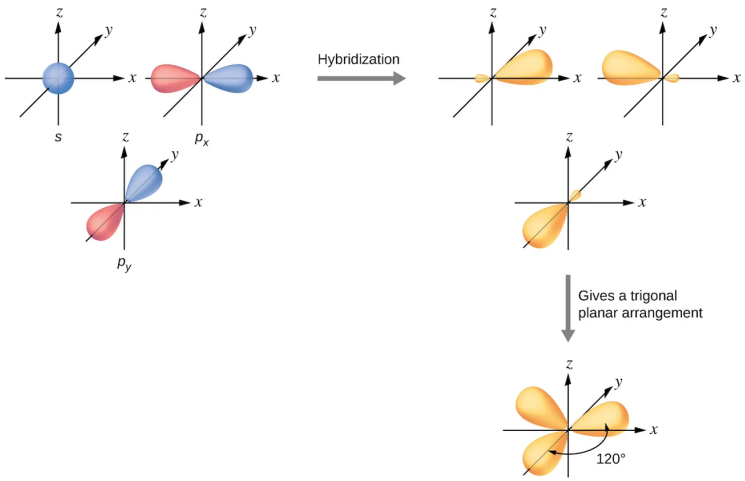


Figure 8.10 The hybridization of an s orbital (blue) and two p orbitals (red) produces three equivalent sp2 hybridized orbitals (yellow) oriented at 120° with respect to each other. The remaining unhybridized p orbital is not shown here, but is located along the z axis.

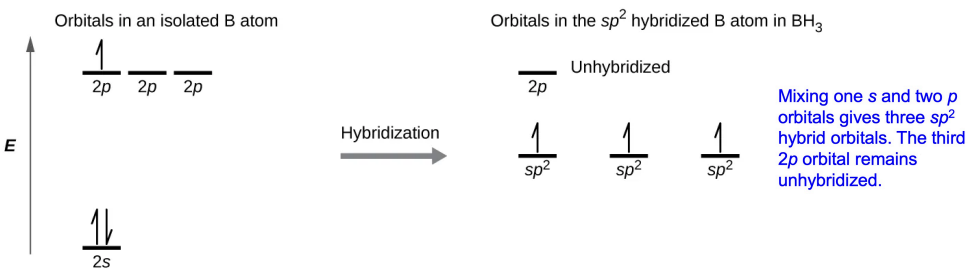
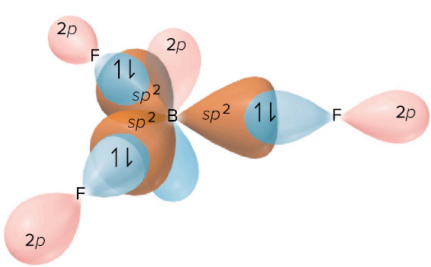


Figure 8.13 In an isolated B atom, there are one 2s and three 2p valence orbitals. When boron is in a molecule with three regions of electron density, three of the orbitals hybridize and create a set of three sp2 orbitals and one unhybridized 2p orbital. The three half-filled hybrid orbitals each overlap with an orbital from a hydrogen atom to form three σ bonds in BH_3.

The three sp^2 orbitals point to the corners of an equilateral triangle, their axes 120° apart. Each half-filled sp^2 orbital overlaps with the half-filled 2*p* orbital of a F atom.



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(c) sp^3 hybrid orbitals in CH_4.

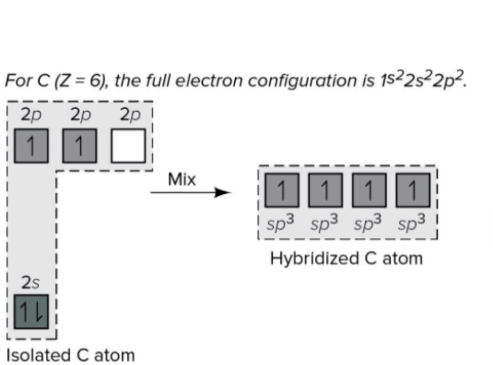
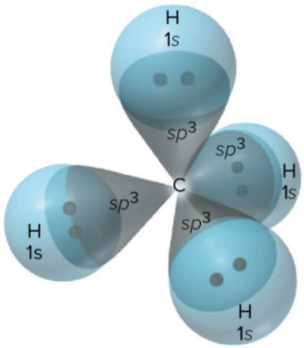
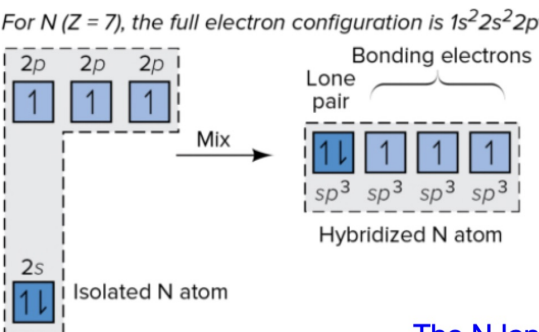


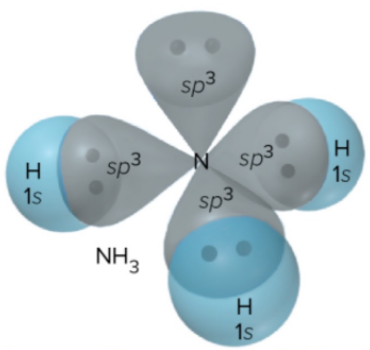
Figure 10.5



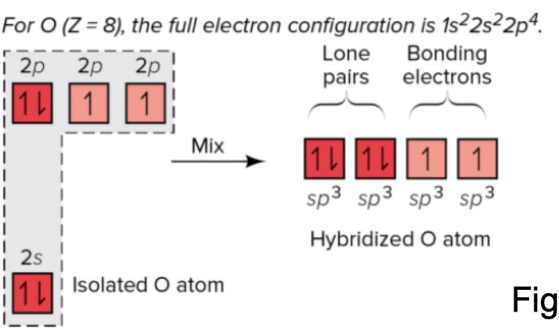
The four sp^3 orbitals adopt a tetrahedral shape.

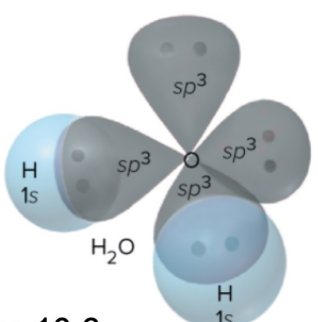
(d) sp^3 hybrid orbitals in NH_3 and H_2O





The N lone pair occupies an sp^3 hybrid orbital, giving trigonal pyramidal shape.



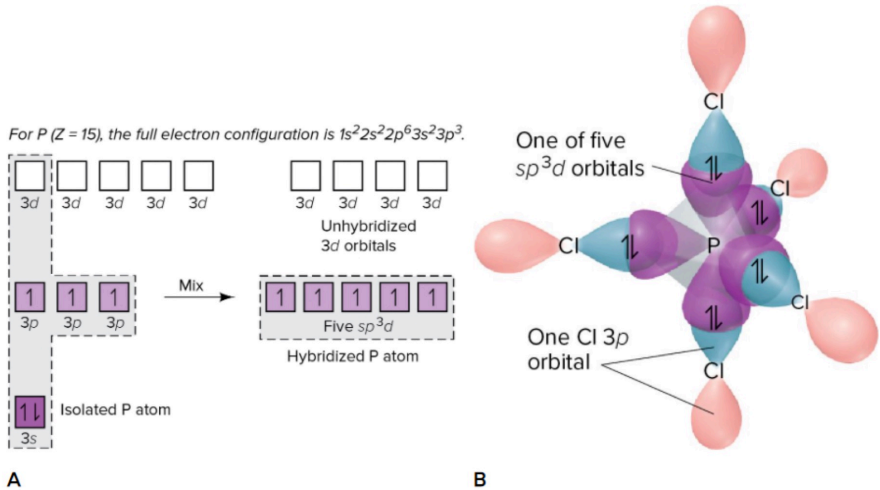


The O lone pairs occupy sp^3 hybrid orbitals, giving a bent shape.

Figure 10.6

(e)

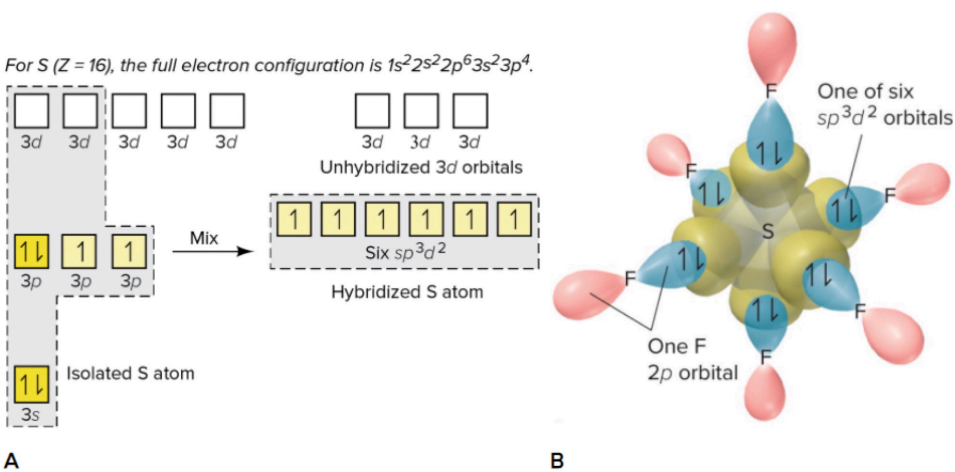
Figure 10.7 The sp^3d hybrid orbitals in PCl_5.



The formation of more than four bonding orbitals requires *d* orbital involvement in hybridization.

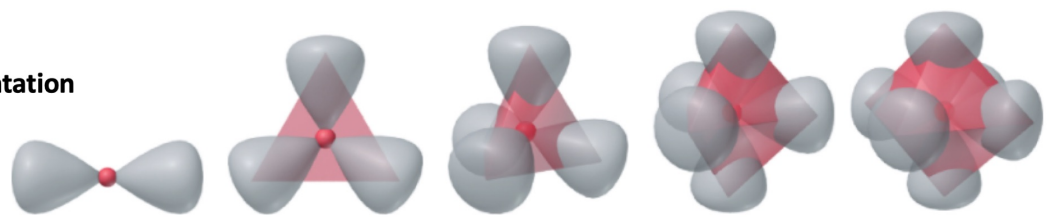
(f)

Figure 10.8 The sp^3d^2 hybrid orbitals in SF_6.



The orbital box diagram shows the formation of six half-filled sp_3d_2 orbitals; three 3*d* orbitals remain unhybridized and empty.

TABLE 10.1 Composition and Orientation of Hybrid Orbitals

	Linear	Trigonal Planar	Tetrahedral	Trigonal Bipyramidal	Octahedral
Atomic orbitals mixed	one s one p	one s two p	one s three p	one s three p one d	one s three p two d
Hybrid orbitals formed	two sp	three sp^2	four sp^3	five sp^3d	six sp^3d^2
Unhybridized orbitals remaining	two p	one p	none	four d	three d
Orientation					

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Methanol, CH_3OH

Be able to sketch and label the orbitals involved in bonding in this molecule.

3. Multiple Bonding (8.3)

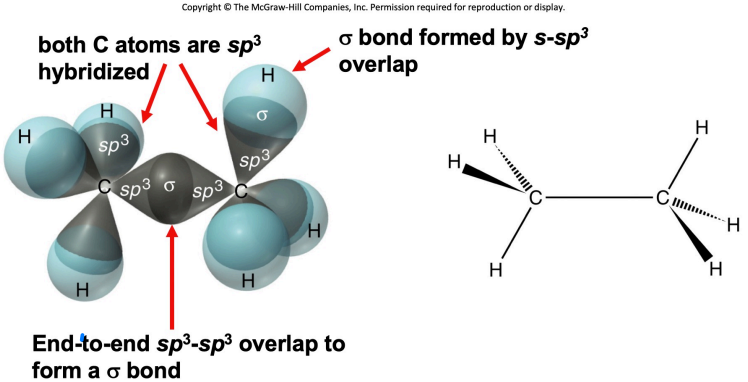
(a) ethane (CH_3CH_3)

- **All bonds in ethane are formed by the direct overlap of orbitals. (Orbitals overlap end-to-end).**
- **sigma(σ) bonds: bonds formed by end-to-end overlap of orbitals. Electron density lies along the bond axis.**

--ALL single bonds are sigma bonds.

Figure 10.10 **The σ bonds in ethane (C_2H_6).**

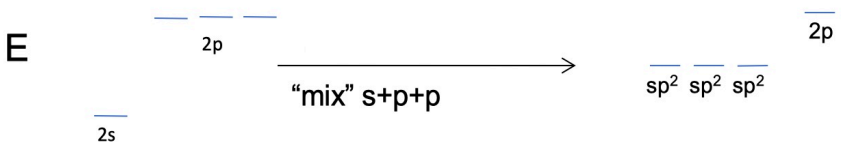
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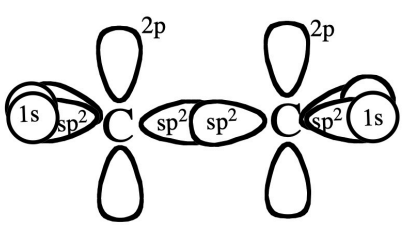
A σ bond is cylindrically symmetrical, with its highest electron density along the bond axis.

(b) ethylene (CH_2CH_2)

Remember: to form sp^2 hybrid orbitals:



In addition to the 5 σ bonds there is a pi (π) bond formed by the overlap of 2p orbitals that remain.



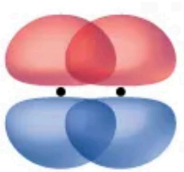


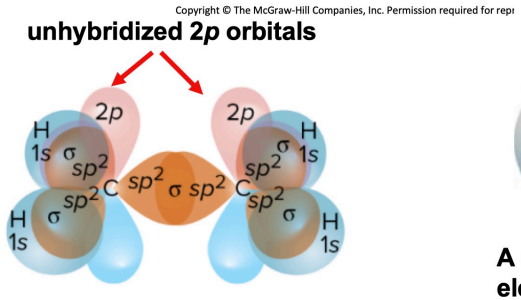
Figure 8.5 Pi (π) bonds form from the side-by-side overlap of two *p* orbitals. The dots indicate the location of the nuclei.

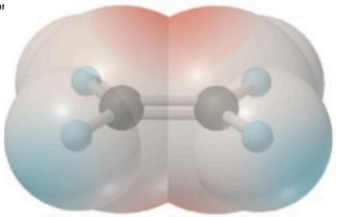
pi (π) bond= bond formed by sideways overlap of two atomic orbitals that has two regions of electron density, one above and one below the σ bond axis. One π bond holds two electrons that occupy both regions (lobes) of the bond.

- **σ bond allows free rotation**
- **π bond restricts rotation (p-orbitals have to be parallel to engage in side-to-side overlap).**

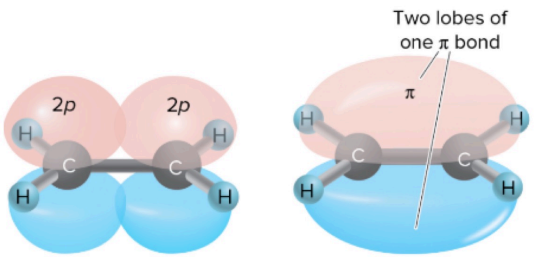
Figure 10.11 The σ and π bonds in ethylene (C_2H_4).

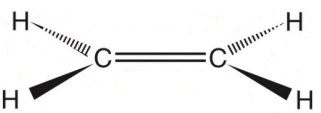
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A π bond has two regions of electron density.

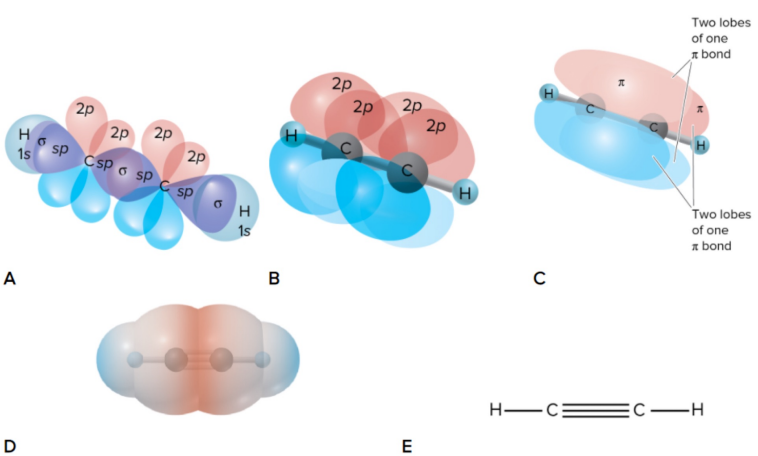




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(c) acetylene (CHCH)

Figure 10.12 The σ and π bonds in ethyne (C_2H_2).



Example: Diagram and describe the bonding in CH_3COOH (acetic acid)

F. Molecular Orbital (MO) Theory (Section 8.4)

• Another model to describe bonding.
• MO's are results of quantum mechanical treatment of molecules.
• Electrons occupy orbitals in molecules similar to how they occupy atomic orbitals, but MO's extend over the molecule. Electrons are <u>delocalized</u>.

Valence Bond Theory	**Molecular Orbital Theory**
considers bonds as localized between one pair of atoms	considers electrons delocalized throughout the entire molecule
creates bonds from overlap of atomic orbitals (*s, p, d*…) and hybrid orbitals (*sp, sp*^{2}, *sp*^{3}…)	combines atomic orbitals to form molecular orbitals (σ, σ*, π, π*)
forms σ or π bonds	creates bonding and antibonding interactions based on which orbitals are filled
predicts molecular shape based on the number of regions of electron density	predicts the arrangement of electrons in molecules
needs multiple structures to describe resonance	

Simple MO's: use atomic orbitals on atoms as a basis for the MO's.

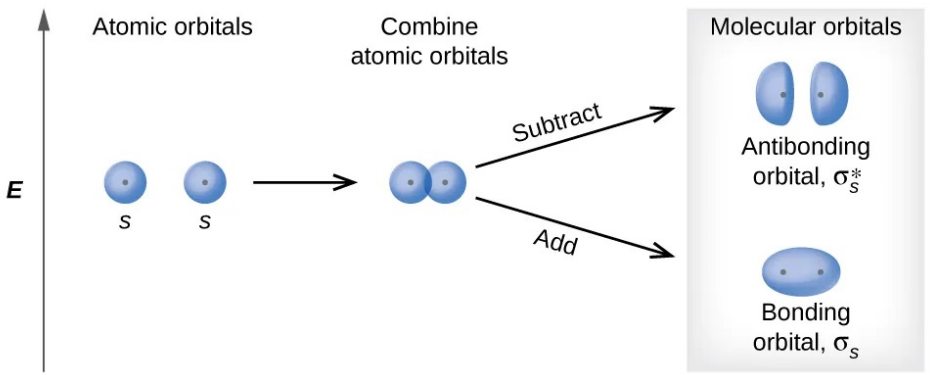
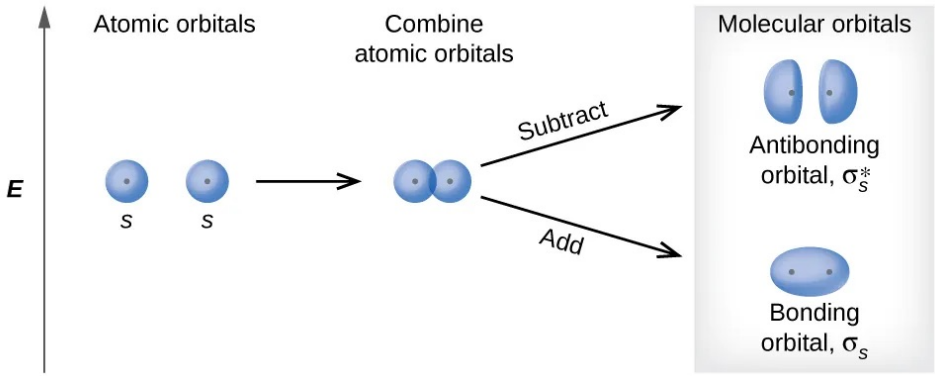


Figure 8.29 Sigma (σ) and sigma-star (σ*) molecular orbitals are formed by the combination of two s atomic orbitals. The dots (·) indicate the locations of nuclei.



The electron probability is cylindrically symmetric along the bond axis: <u>sigma (σ) MO's</u>.

The name represents the symmetry, the parent AO, and whether the MO is bonding or antibonding: σ_{1s} **and** σ_{1s}^*

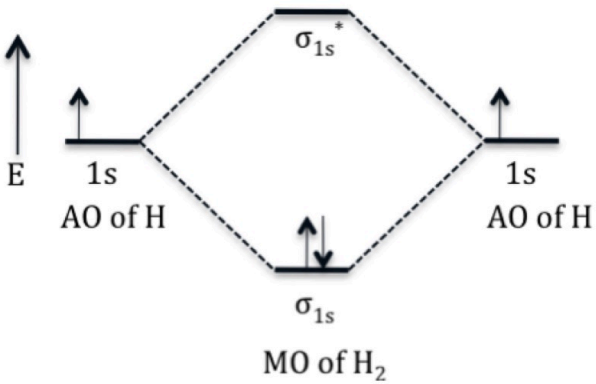
σ_{1s}—greatest electron probability is between nuclei.

- Electron-nucleus attraction.
- Electrons can delocalize charge over larger volume than original AO's, thus e-/e- repulsions are reduced.
- Lower energy.

σ_{1s}^*

- node between nuclei.
- Electrons do not shield one nucleus from the other--nucleus/nucleus repulsion
- Higher energy.

MO DIAGRAM for H_2



We can write a ground state valence electron configuration for the molecule:

Bond Order = ½ [(# of electrons in bonding MO's) – (# of electrons in antibonding MO's)]

- A **bond order > 0** means that the molecular species is stable relative to separated atoms.
- **Bond order = 0** means no net stability.

A similar MO diagram can be drawn for He_2

You will find that:

Bond order = ½(2-2) = 0 Unstable (does not exist)

Configuration is: $(\sigma_{1s})^2(\sigma_{1s}^*)^2$

MO DIAGRAM for Li_2

Combination of atomic p-orbitals combine to form MO's

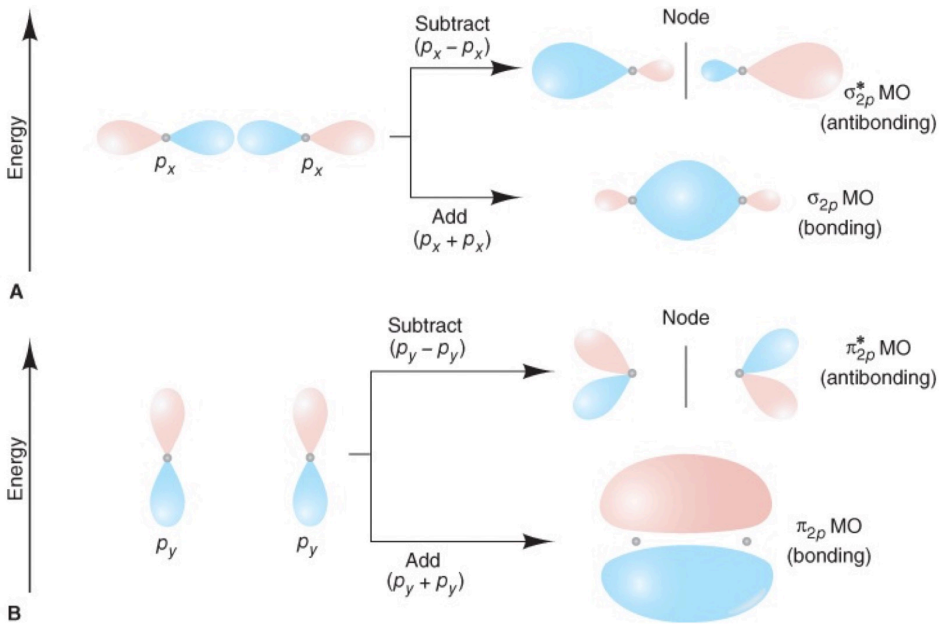


Figure 10.21

Shapes and energies of σ and π MOs from combinations of 2*p* atomic orbitals.

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MO diagram for O_2

Expected MO energy diagram

(true for O_2, F_2, Ne_2)

In B_2 , C_2, N_2:

- **The MO energy order is not as expected.**
- **The 2s AO on one atom interacts with the 2p atomic orbital on the other.**
- **Interaction causes the σ_{2s} MO to have lower energy and the σ_{2p} MO to have higher energy.**

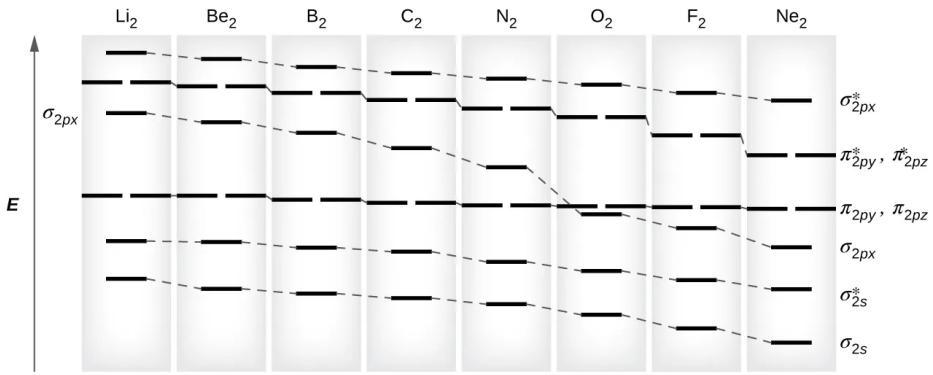
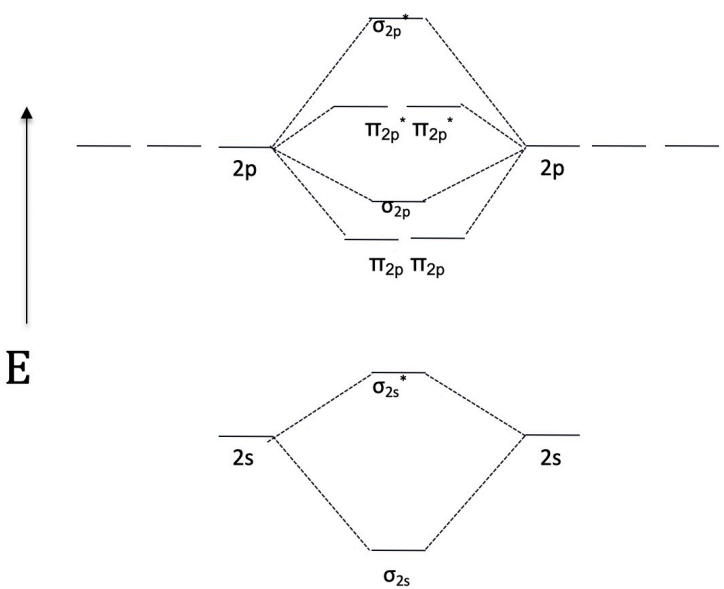


Figure 8.37 This shows the MO diagrams for each homonuclear diatomic molecule in the second period. The orbital energies decrease across the period as the effective nuclear charge increases and atomic radius decreases. Between N2 and O2, the order of the orbitals changes.



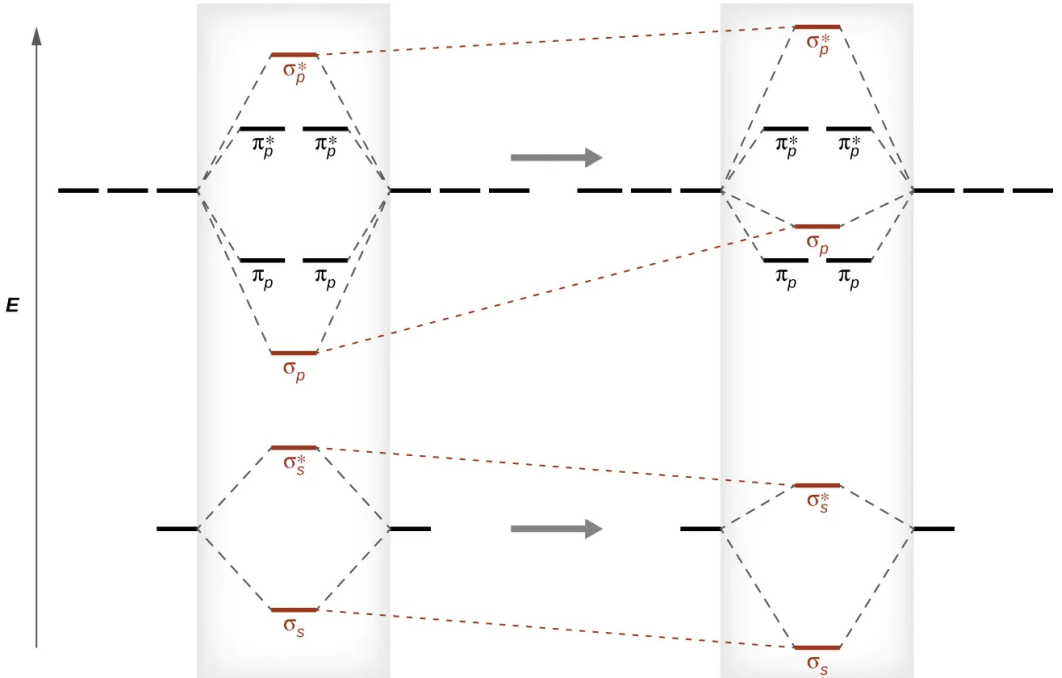
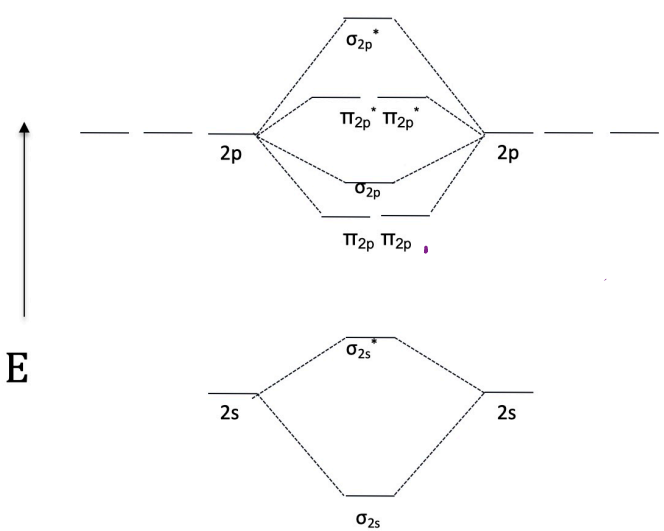


Figure 8.38 Without mixing, the MO pattern occurs as expected, with the σp orbital lower in energy than the πp orbitals. When s-p mixing occurs, the orbitals shift as shown, with the σp orbital higher in energy than the πp orbitals.

MO DIAGRAM for B_2



MO for Diatomic ions: Example N_2^-

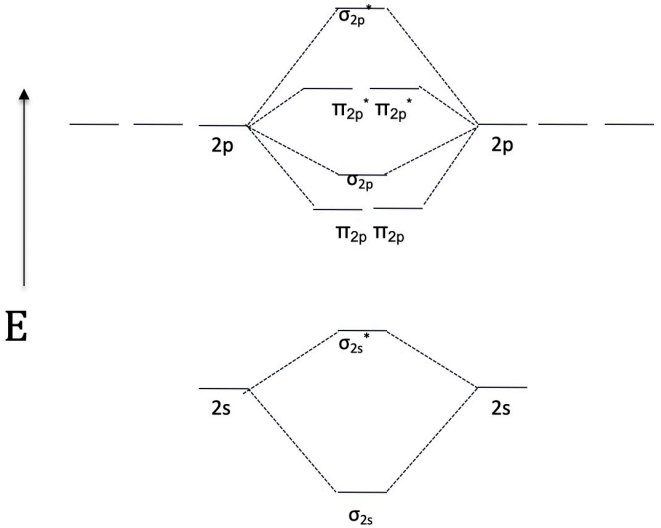
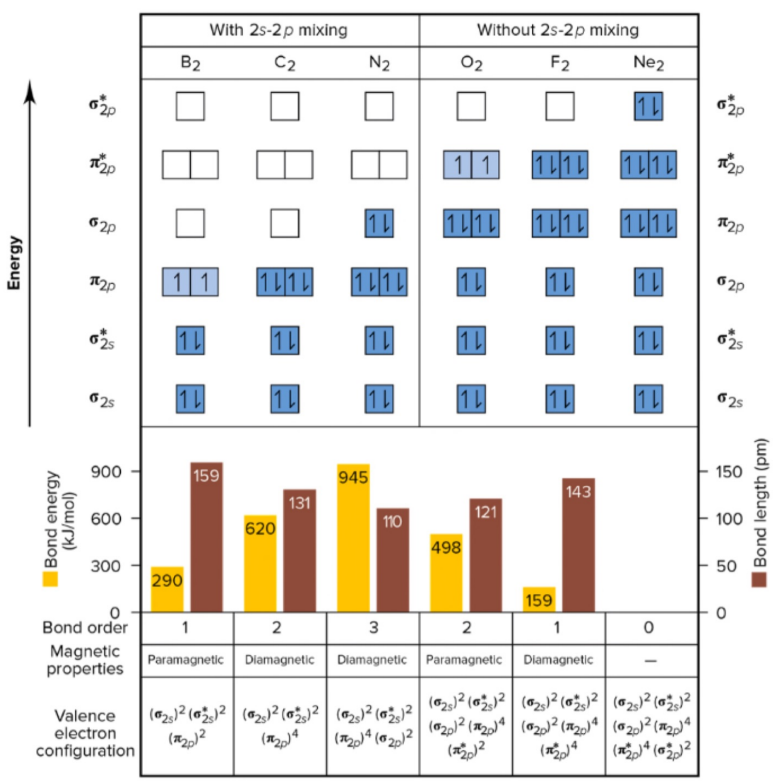


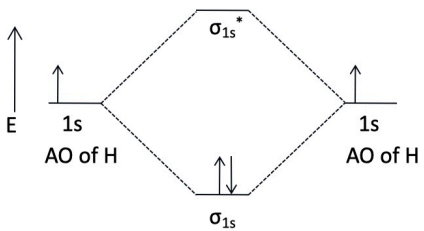
Figure 10.23
MO occupancy and some properties of homonuclear diatomic molecular properties for B_2 through Ne_2.



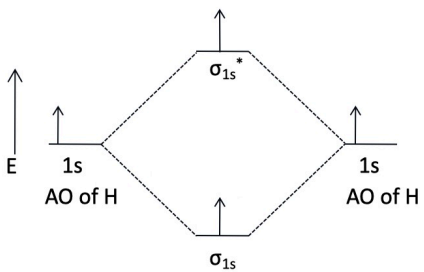
MO Diagrams for Excited State Molecules and Ions

Atoms:	
n = 1	ground state
n = 2	first excited state
n = 3	second excited state
⋮	
⋮	
n = ∞	ionization

Molecules:
Electrons normally **fill the lower energy MOs = Ground state**.
An electron can be promoted to a higher energy MO by absorption of energy = excited state of the molecule.



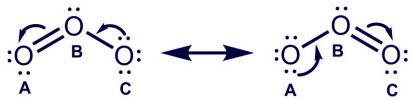
MO of H_2



MO of H_2^*

Valence bond theory treats all bonds as localized, whereas MO theory treats all bonds as completely delocalized. Compare how these theories explain bonding in ozone (O_3) and benzene, C_6H_6.

Valence bond theory:
O_3 can be drawn in 2 ways:



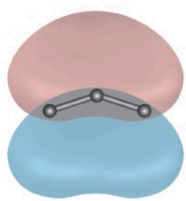
O_3 can be depicted by more than one valid Lewis structure and is best represented as a ***resonance hybrid***.

Resonance forms ***are not real bonding depictions***.

O_3 does ***not*** change back and forth between its two resonance forms.

The ***real structure*** of the resonance hybrid for O_3 is an ***average*** of its contributing resonance forms.

Molecular Orbital Diagram of lowest energy π-bonding MO in ozone: delocalized π bonding electron pair over entire molecule.



Ozone, O_3

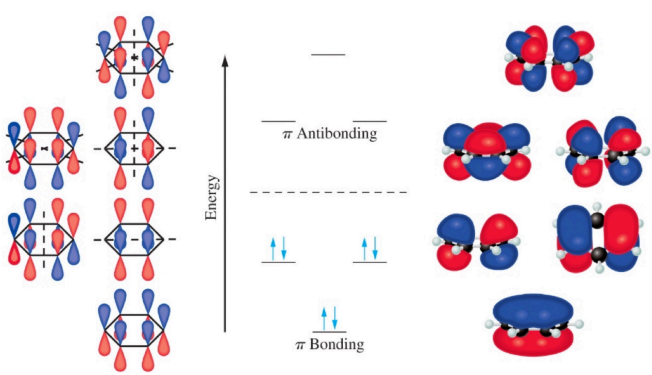
Representations of benzene.

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Resonance forms having alternating single and double bonds.

Resonance hybrid shows the delocalized electrons as either an unbroken or a dashed circle.



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